

Trainings ▾

General Information

Industrial Applications

Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance checks that are due for scheduled maintenance.

Maintenance Procedures at Daily Interval (Section 3)

- Drive Belts - Check
- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Fan, Cooling - Check
- Coolant Level - Check
- Charge Air Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain

Maintenance Procedures at 250 Hours, or 6 Months¹ (Section 4)

- Fuel Filter (Spin-On Type) - Change
- Lubricating Oil and Filters² - Change
- Coolant Filter³ - Change
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check
- Engine Wiring Harness - Check

Maintenance Procedures at 1500 Hours, or 1 Year¹ (Section 5)

- Engine Steam Cleaning - Clean
- Overhead Set - Adjust
- Radiator Shutter Assembly - Check
- Water Pump - Check
- Cooling System Hoses - Check
- Air Cleaner Assembly (Engine-Mounted) - Check
- Air Leaks, Air Intake and Exhaust Systems - Check
- Turbocharger - Check
- Air Cleaner Restriction - Check
- Batteries - Check
- Engine Mounts - Inspect for Reuse

Maintenance Procedures at 6000 Hours, or 2 Years¹ (Section 6)

- Vibration Damper, Viscous - Check

- Cooling System - Flush
- Fan Drive Idler Pulley Assembly - Check
- Fan Hub, Belt Driven - Check
- Turbocharger - Check
- Air Compressor Discharge Lines - Check

Reference the appropriate sections for complete inspection and maintenance procedures.

1. These maintenance intervals **must** be performed at hours or months, whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance procedures that are due for scheduled maintenance.
2. Fuel burn is the most accurate method for determining oil change intervals. Oil change intervals can be required more frequently, depending on the quality of the oil available. Reference the Oil Drain Interval table for marine and industrial engines in this procedure for additional oil change intervals.
3. Do **not** change the coolant filter if the SCA concentration level is over three units.

Oil Drain Intervals

QSM11 Industrial Engines⁽⁵⁾ - Oil Drain Intervals by Duty Cycle (hours)⁽²⁾

	Severe	Heavy	Medium	Light
Fuel Consumption ⁽⁴⁾	Greater than 13 gallon/hour	10 to 13 gallon/hour	8.5 to 10 gallon/hour	Less than 8.5 gallon/hour
API CF-4 ⁽³⁾	150	200	250	300
API CG-4	250	300	400	500
API CH-4 or ACEA E5	300	350	500	600
CES20076 ⁽¹⁾ or CES20077	350	400	550	700

Notes:

1. Valvoline™ Premium Blue™ and Premium Blue™ 2000 meet 20076 standards.
2. Limit oil change interval to 250-hour maximum for all oil grades if 40 percent of the operating time is in ambient temperatures above 38°C [100°F] to prevent engine damage due to oil degradation resulting from high oil temperatures.
3. Cummins Inc. does **not** recommend the use of CE or CD grade oils. If CD or CE oils **must** be used, make the oil drain intervals half of the API CF-4 values.
4. Engine oil drain intervals need to be reduced by 50 percent when fuel sulfur content exceeds 0.5 percent.
5. Oil filters **must** meet CES 10065 specifications. LF9001 meets and exceeds CES 10065 specifications.

